

Programme Code: TU856
Shared with: DT844
Module Code: CMPU 3051
CRN: 30102

TECHNOLOGICAL UNIVERSITY DUBLIN

City Campus - Grangegorman

TU856 - BSc. Degree in Computer Science

Year 3

SEMESTER 1 EXAMINATIONS 2024/2025

CMPU3051 Introduction to Artificial Intelligence

Internal Examiner(s):

Mr. Richard Lawlor
Dr. Paul Doyle

External Examiner:

Dr. Colm O'Riordan

Exam Duration:

2 hours

Instructions To Candidates:

Attempt 4 out of 5 questions. All questions carry equal marks.

Special Instructions /Handouts/ Materials Required:

None

1 (a) Explain what is meant by *state space search*.

In state space search, discuss the difference in performance between depth first and breadth first search. Assume each state has **b** successor or branching states. Is there a way to combine the advantages of both?

(9 marks)

(b) Represent the following statements in Predicate Calculus:

- Lucy is a professor
- All professors are people
- John is the dean
- Deans are professors
- All professors consider the dean a friend or don't know him
- Everyone is a friend of someone

(8 marks)

(c) Prolog uses backward chaining. Explain what this means with the help of a proof tree in terms of the following Prolog program when the Prolog interpreter is asked to prove **k(x)**. Make reference to *unification* in your answer.

f(a) .

f(b) .

g(a) .

g(b) .

h(b) .

k(X) :- f(X), g(X), h(X) .

(8 marks)

2 (a) Write down 2 versions of Bayes' rule.

Given the following:

1% of people have a certain genetic defect.

90% of tests for the gene detect the defect (true positives).

9.6% of the tests are false positives.

Use Bayes' rule to calculate the odds that person actually has the genetic defect if they get a positive test result.

(9 marks)

- (b)** Give three reasons why Bayesian reasoning is difficult to use in an expert system. Also, briefly describe two advantages in using certainty factors as opposed to Bayesian reasoning.

Briefly mention two well-known historical expert systems, one of which used Bayesian reasoning and the other which used certainty factors.

(8 marks)

- (c)** Given the forecasting rules below, show how they would fire to forecast tomorrow's weather when provided with the following information: there is rain today and the rainfall is low with a certainty factor of 0.8, and also it is cold with a certainty factor of 0.9. This formula may be useful:

$$cf(cf_1, cf_2) = cf_1 + cf_2 \times (1 - cf_1)$$

Rule: 1

if today is rain

then tomorrow is rain {cf 0.5}

Rule: 2

if today is dry

then tomorrow is dry {cf 0.5}

Rule: 3

if today is rain

and rainfall is low

then tomorrow is dry {cf 0.6}

Rule: 4

if today is rain

and rainfall is low

and temperature is cold

then tomorrow is dry {cf 0.7}

Rule: 5

if today is dry

and temperature is warm

then tomorrow is rain {cf 0.65}

Rule: 6

if today is dry

and temperature is warm

and sky is overcast

then tomorrow is rain {cf 0.55}

(8 marks)

- 3 (a) Express the following Predicate Calculus function and statements in Prolog:

`minincome(Y) = 15000 + 4000 * Y`

`savings_account(adequate) ∧ income(adequate) ⇒ investment(stocks).`

$\forall X \text{ earnings}(X, \text{steady}) \wedge \forall Y \text{ dependents}(Y) \wedge \text{greater}(X, \text{minincome}(Y))$
 $\Rightarrow \text{income(adequate)}.$

(5 marks)

- (b) Write a recursive Prolog predicate **canGoFromTo(A, B)** to express the following:
One can travel from A to B if there is a bus or train from A to B. Or one can travel from A to B if there is a bus or train from A to C and one can travel from C to B.

(5 marks)

- (c) Write a Prolog list which represents the state of the 8-puzzle shown here:

1	5	3
8		2
7	4	6

(5 marks)

- (d) Complete the Python code below for the training operation for a single perceptron with binary output. You can assume each row in **xs** is an input vector and **ys** is a vector with the corresponding desired outputs.

```
class Perceptron(object):
    def __init__(self, n):
        self.n = n
        self.w = np.random.randn(n,)
        self.b = np.random.randn()

    def train(self, xs, ys, epochs, eta):
        cost = np.zeros((epochs,), dtype=float)
        for ep in range(epochs):
```

(5 marks)

- (e) A simple perceptron is required which decides on whether or not to buy an expensive cocktail based on the following inputs where $x = [x_1, x_2, x_3]$.

$x_1 = 1$ means you have plenty of cash, $x_1 = 0$ means you do not. $x_2 = 1$ means you want to impress a new friend, 0 means you don't. $x_3 = 1$ means it's a busy exciting night on the town, 0 means it is not.

If you want to impress someone, you will buy cocktails no matter what. However, if you do not wish to impress but it's an exciting night and you have plenty of cash, you will buy a cocktail. Otherwise you will not buy a cocktail.

Using the class outlined in part(b), write code that creates a perceptron object to for the problem described above and which then calls the training operation on it with appropriate parameters. You will need to to define the input vectors and the desired output values.

(5 marks)

- 4 (a) Draw a diagram for a single two-input perceptron. Explain what is meant by linear separability. How many categories can it classify inputs into? Mention any issues involved in using a perceptron to compute logical AND, OR and XOR. (5 marks)
- (b) Illustrate the single perceptron training algorithm for an OR gate by computing the weight adjustments for the first two epochs in Table 4(b) below. (5 marks)
- (c) Distinguish between a sigmoid neuron and a perceptron, and outline the difference in the training formula for Δw_i .
- Why might it be preferable to use sigmoid neurons in a neural network rather than perceptrons? (5 marks)
- (d) Suppose you intend to use a 3-layer ANN (an input layer, one hidden layer and an output layer) and the weight matrices are $\mathbf{w}_2$ for the hidden layer and $\mathbf{w}_3$ for the output layer and the bias vectors are $\mathbf{b}_2$ and $\mathbf{b}_3$.

Given an input column vector $\mathbf{x}$ and a corresponding target output vector $\mathbf{y}$, write some Python code which:

- computes the output vectors $\mathbf{a}_2$ and $\mathbf{a}_3$ for layer 2 and 3 respectively assuming a sigmoid activation function;
- computes the cost for the current input assuming the mean-squared-error cost function;
- computes $\mathbf{\delta}_3$ for the output layer and $\mathbf{\delta}_2$ for the hidden layer.

(10 marks)

Perceptron Training for logical OR

Threshold $\theta = 0.2$, learning rate $\alpha = 0.1$.

Epoch	Inputs		Desired output Y_d	Weights		Weighted Sum z	Actual Output a	Error e	Weight adjustments	
	x_1	x_2		w_1	w_2				Δw_1	Δw_2
1	0	0	0	0.3	-0.4					
	0	1	1							
	1	0	1							
	1	1	1							
2	0	0	0							
	0	1	1							
	1	0	1							
	1	1	1							
3	0	0	0							

Table 4(b)

- 5 (a) How is a state described in the STRIPS AI planning system?

For the *Monkey and Banana* problem, express in Prolog a possible initial state and a goal state that would be appropriate for STRIPS.

(5 marks)

- (b) How is a STRIPS operation described?

Outline two STRIPS operations that would be suitable for solving the *Monkey and Banana* problem and then write Prolog code for each of them.

(10 marks)

- (c) Show how the STRIPS search algorithm could be written in Prolog.

What is the main disadvantage of STRIPS?

(10 marks)